

Automatic Modulation Classification: A Novel Convolutional Neural Network Based Approach

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Abstract—Deep learning (DL) is a new paradigm of machine learning (ML) that has shown exceptional performance in image, voice and natural language processing. However, researchers have not explored the use of DL to wireless communication to its full potential. The use of DL technology for wireless communication applications has recently gained popularity. This paper looks into the application of deep learning based approach for automatic modulation classification (AMC). Automatic modulation classification has a diverse applications ranging from civilian to military. A deep learning based convolutional neural network (CNN) architecture for AMC is proposed in this paper. We make use of Gaussian noise layer after convolution layers in our proposed architecture which has a regularization effect while training and it reduces over fitting problem. We demonstrate using experiments that the proposed architecture outperforms the existing CNN based architectures for AMC. We also demonstrate the effects of different architecture parameters on the performance of the proposed algorithm.

Index Terms—Deep learning, Machine Learning, Automatic Modulation Classification, Convolutional Neural Network

I. INTRODUCTION

The capability to quickly learn and label the radio frequency (RF) spectrum autonomously serves as a critical enabling factor for monitoring spectrum interference, detecting radio fault, dynamic spectrum access (DSA), and various defense and regulatory applications. Automatic modulation classification (AMC) was first proposed for military applications, where it was used to recognize signal modulations to identify adversary transmitting units, prepare jamming signals for enemy transmissions, and recover the intercepted enemy signals. The carefully hand-crafted specialized feature extractor for specific signal types is traditionally used for radio signal classification, and modulation recognition [1]. The classification task is carried out using analytically derived decision boundaries, or the decision boundaries learned statistically. However, we have recently seen rapid developments in deep learning (DL) architectures, replacing state-of-the-art machine learning algorithms in image, voice, and natural language processing. The application of recent advances in Deep Learning [2] to wireless communications has shown the possibility of achieving extraordinary performance improve-

ments that would otherwise be impossible to achieve through more traditional approaches.

A modulation classifier's design involves the following steps: signal pre-processing and building/selecting a proper algorithm for classification. For preprocessing of the data, signal processing techniques such as filtering can be used for noise reduction. The preprocessing step may also include estimation of frequency, time period and signal power. The channel equalization techniques can be applied on the data to preprocess the data for reducing the channel effects. The classification algorithm in second step can be based on likelihood-based approach [3] [4] or a feature-based approach [5] [6] [7]. In the likelihood-based method, the likelihood ratio is compared against a properly chosen threshold such that the probability of false classification is minimized. On the other hand, in the feature-based approach, a decision is made based on the observed values of several features. Our model involves using a feature-based process that is extracted using different layers involved in deep learning architecture. The proposed research work is mainly oriented towards designing new deep learning based convolutional neural network (CNN) algorithm for automatic modulation classification. In the following subsection, we discuss few works based on CNN architecture for AMC [8], [9].

A. RELATED WORK

The Visual tree convolution neural network 2 (VTCNN2) system consisting of convolution, and dense neural network layers is proposed in [10]. The pooling and regularization layers are not utilized in VTCNN2. The pooling layer is used to progressively reduce the size of the representation to minimize the computational cost. The regularization is used to reduce the over-fitting issue in the deep learning architectures to improve the test accuracy.

The model proposed in [11] uses an artificial neural network (ANN) with a varying number of dense layers. A further pre-processing such as feature extraction and pooling is required that will improve the performance significantly. The architecture proposed in [12] uses convolution layers and two fully connected dense layers with dropout. However, this architecture has made significant improvement in accuracy and computation time, but the addition of noise and pooling

layer could have made it more robust. The authors in [12] have done a considerable analysis on convolution neural network (CNN) based architecture and show that it is ideal for improving automatic modulation classification.

The research carried out by authors in [13] is based on different deep learning architectures for AMC have made a profound impact in this field. They have proposed four different architectures as follows: seven staged CNN, DenseNet, ResNet and, CLDNN. Furthermore, it consists of convolution layers linked to fully connected dense layers and dropout at different points to provide regularization. However, the CLDNN algorithm contains long-term memory (LSTM) layer connected between convolution and dense layers providing the improved accuracy.

In this work, we propose novel CNN-based architecture for AMC. We carry out the comparison of the proposed algorithm with the existing CNN based architectures. The proposed architecture outperforms the existing state-of-the-art CNN based architectures.

II. PROBLEM STATEMENT

In general, the modulation classification is a class classification problem with N classes. The general form of received wireless signal at the receiver can be represented as

$$y(t) = c(t) * x(t) + n(t), \quad (1)$$

where $x(t)$ represents the complex baseband envelop of the received signal (noise free), $n(t)$ represents the Additive White Gaussian Noise (AWGN) having distribution $\mathcal{N}(0, \sigma_n^2)$ and $c(t)$ is the impulse response (time varying) of a channel between transmitter and receiver. The modulation classifier's basic aim is to decide based on the only observation $y(t)$ the class an $x(t)$ belongs to. In mathematical terms the classifier gives out the probability $P[x(t) \in N_i | y(t)]$ where N_i represents the i^{th} class. Due to the flexibility and simplicity of mathematical operations, the IQ format is commonly used to represent the received signal. Let A be the instantaneous amplitude and ϕ be the phase of the received signal $x(t)$. Using this, the in-phase and quadrature components can be represented as $I = A \cos(\phi)$ and $Q = A \sin(\phi)$, respectively.

The RadioML dataset discussed in the next section is used for training and testing of the proposed model. The signal expression given in Eq. (1) is used for creation of the dataset.

III. DATASET DESCRIPTION

In this section, we give details of the publicly available RadioML dataset used in this work for evaluating the performance of the proposed model.

A. SOURCE

The dataset RADIOML 2016.04.C is an open source dataset which can be downloaded from <https://www.deepsig.ai/datasets>. It is a dataset with variable SNR included with practical imperfections such as moderate local oscillator drift and fading. The dataset includes signal samples at different labeled SNR increments that can be used to measure the performance across different SNR scenarios.

TABLE I: Table of Notations

Modulation type	AM-DSB, AM-SSB, BPSK, CPFSK, GFSK, PAM4, QAM64, QAM16, QPSK, WBFM, 8PSK
Length of each sample	128
Range of SNR	-20 dB to +18 dB
Training samples	90753
Validation samples	38895
Test Samples	32412

B. GENERATION

The dataset was generated by combing the logical modules together in a GNU Radio as mentioned in [10]. The process of generation starts with selecting a voice signal for analog signals and ASCII symbols for digital modulation. The second step involves the normalization of steps per symbol to form a constant rate of symbols before modulation and applying modulation to the signal. The final step is channel simulation which is designed using the dynamic channel model block in GNU Radio. This block can be used to includes different channel effects such as random processes for center frequency offset, sample rate offset, AWGN, multi-path, and fading. This process generates a dataset which different classes of modulation at different SNR values.

C. DETAILS

The dataset consists of 1,62,060 rows categorized in three columns: SNR, modulation type, and data of the signal. It includes a comprehensive range of SNR values starting from -20dB to +18dB, progressing at an interval of 2dB. Furthermore, the dataset is classified into 11 different types of modulation, of which eight are digital, and three are analog. The modulation types and the details of dataset parameters are tabulated in TABLE I. The signal data (complex-valued inputs) for each SNR and modulation type consists of an array of dimensions [2, 128] where the orthogonal synchronously sampled I & Q samples form the 2-wide dimension.

D. VISUALIZATION

Fig. 1 shows visualization for different modulated signals at 18 dB SNR. The visualization of this dataset is employed by the time-domain plot of the I and Q values of signal data. The fundamental differences among all the classes can be easily spotted, but the channel effects are not readily visible. We can clearly spot the difference between different modulation shown in Fig. 1 because the plots are obtained at very high SNR values. In case of low SNR, the noise becomes dominant making it difficult to distinguish between different modulations.

IV. PROPOSED ARCHITECTURE

The Fig. 2 shows the detailed architecture of the proposed model. It is divided into six stages which are described as follows:

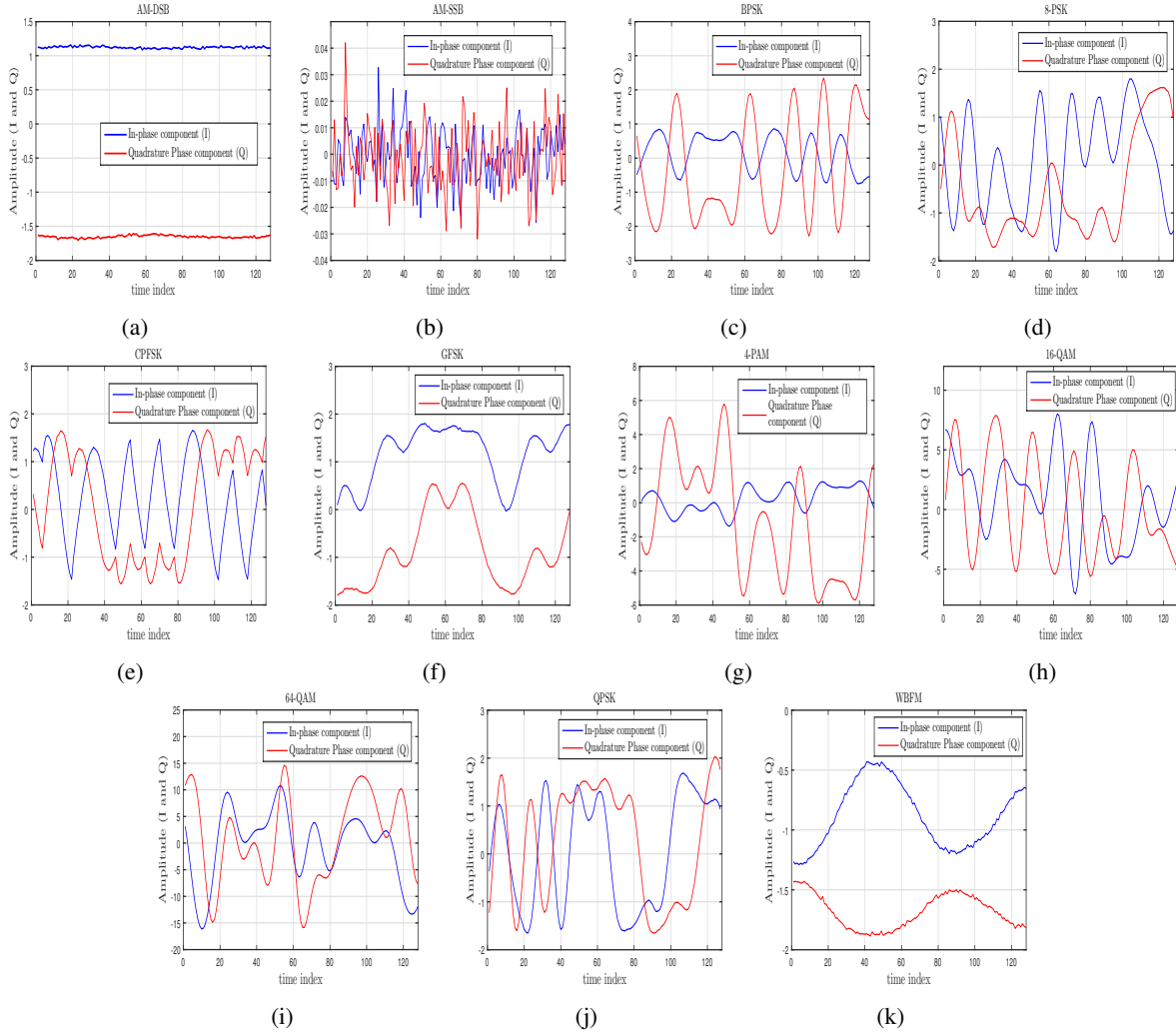


Fig. 1: The in-phase and quadrature phase components for different modulation schemes plotted for SNR=18 dB

- Stage 1: The first stage consists of three layers involving zero-padding to the input array that progresses to the first convolution layer. The first convolution layer uses 256 filters along with a kernel window of (1,2). The activation used at this stage is ReLu and it ends with a dropout layer.
- Stage 2: The second stage consists of a pooling layer that will reduce the training parameter and extract the essential features observed in the dataset. The proposed second stage is the averagepooling2D layer with a kernel size of (1,2), which computes the average and reduces the parameters. This layer is instrumental in increasing the robustness and performance of the model.
- Stage 3: The third stage consists of three layers involving zero-padding (0, 2) to the array that progresses to the second convolution layer. The second convolution layer uses 96 filters along with a kernel window of (1, 4). The activation used at this stage is ReLu and it ends with a dropout layer with a fixed amount.
- Stage 4: The fourth stage consists of three layers involving zero-padding (0, 2) to the array that progresses to

the third convolution layer. The third convolution layer uses 64 filters along with a kernel window of (2, 6). The activation used at this stage is ReLu and it ends with a dropout layer with a fixed value.

- Stage 5: The fifth stage consists of 2 layers. The first layer is used to flatten the output array from convolution to a dimension suitable for the fully stacked perceptron layer. The second layer is the Gaussian noise layer applied as an input to the perceptron. Moreover, the second layer is a crucial part of this entire model, which improves the model's robustness and acts as a regularization layer.
- Stage 6: The sixth stage consists of four fully stacked hidden layers with 1024, 256, 64, and 11 nodes, followed by the SoftMax activation layer for categorical classification.

V. RESULTS

This section of the paper focuses on the performance evaluation of the proposed model. Furthermore, a comparison study is carried out by recreating the algorithms and architecture

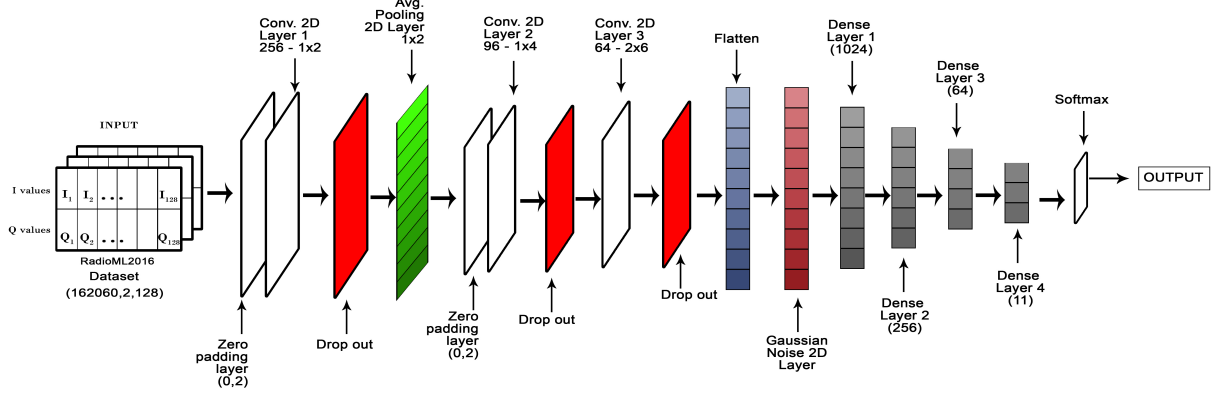


Fig. 2: Proposed CNN based architecture for AMC

mentioned in [10] [11] [12] [13] using RADIOML 2016.04.C.

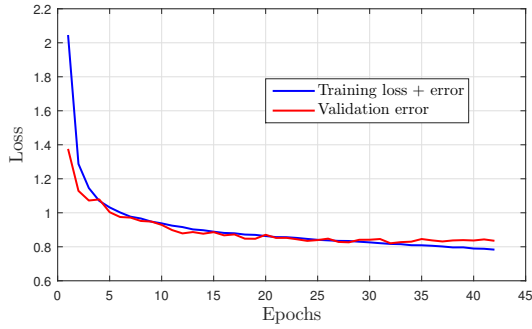


Fig. 3: Loss plot for the CNN architecture

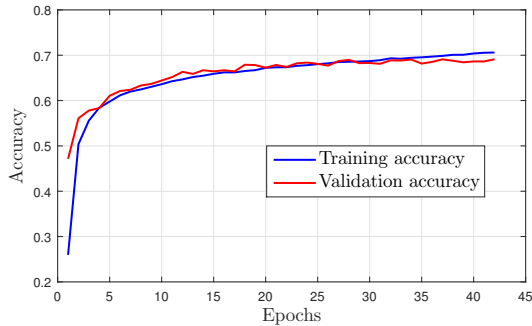


Fig. 4: Accuracy plot for the CNN architecture

A. TRAINING DETAILS

The dataset is split into 80% for training data and 20% for testing data. The ADAM optimizer technique is used for training where the categorical cross-entropy is used as a loss measure. Implementing the model for training and testing takes place on an NVIDIA Cuda enabled GeForce 1080 Ti GPU. The implementation uses an early stopping call-back method that measures the validation set's progress for a number of epochs to reduce over-fitting and roll back to the best model. The model was trained over 90753 data samples that achieved an accuracy of 70.60%, with a loss of 0.7832.

B. VALIDATION DETAILS

The validation split for the best model was selected as 0.3. The training results gave the best validation accuracy of 69.10% and a validation loss of 0.8206. All the results were recreated using our dataset for the model proposed by [10]–[14] provided a validation loss of more than 0.84, which is a vital improvement by the our proposed model.

The Fig. 3 shows the plot for loss versus epochs, and the Fig. 4 shows the plot for accuracy versus epochs for both training and validation data. The presence of regularization methods like adding Gaussian noise and dropout layers is responsible for early convergence, i.e., within 40 to 45 epochs with better results.

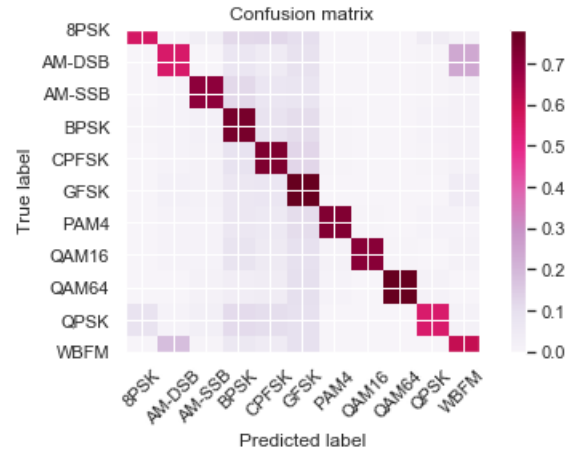


Fig. 5: Confusion matrix of test data for all SNR values

C. TEST DETAILS

The test accuracy obtained by the proposed system is 68.946%, with a loss of 0.8242. The parameters obtained by this architecture is better than the recreated results on [10] [11] [12] [13] that achieved the highest test accuracy of 65.39% with this dataset.

The effect of noise layers in deep learning has a significant impact on performance parameters. It mitigates overfitting and

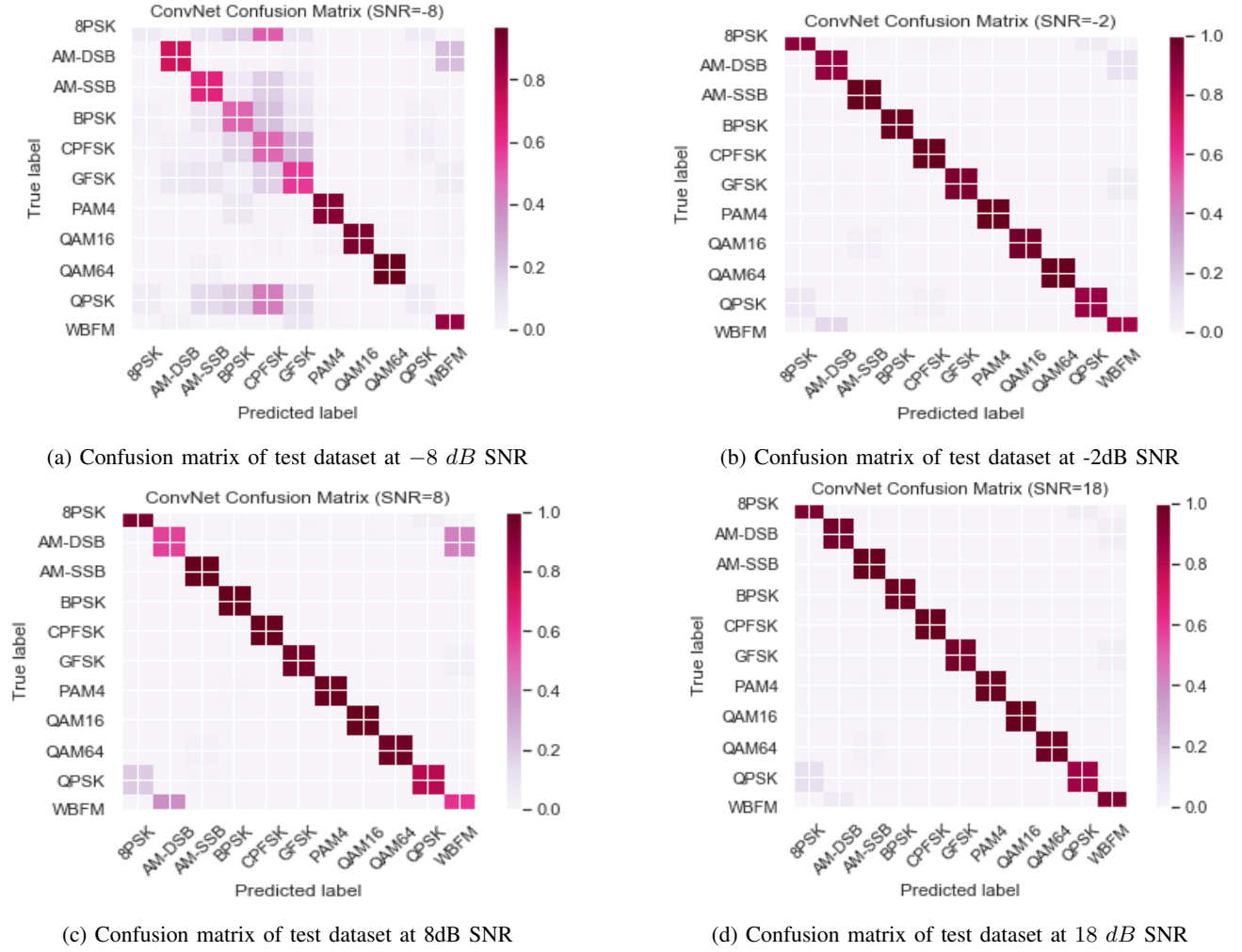


Fig. 6: Confusion matrix of test data at different SNR values (a) SNR= -8 dB, (b) SNR= -2 dB, (c) SNR= 8 dB and (d) SNR= 18 dB

generalizes the data very efficiently to classify. One of the many observations obtained is that the noise layer eliminates the memorization effect and improves the algorithm's learning rate. Based on the paper [15], it was observed that Gaussian noise has the maximum impact of regularization, and the convergence rate is faster with a new minimum loss. All the observations are compared with the models proposed by [10] [11] [12] [13]. Another consideration from [15] is equally applicable to the proposed system because accuracy and convergence rate improves with a complex deep neural network.

D. CONFUSION MATRIX AND SNR PLOT DETAILS

The Fig. 5 shows the confusion matrix visualization of test data for all SNR values. The results highlight that WBFM is misclassified as AM-DSB up to some extent and vice-versa. Moreover, the test result depicts a misclassification among BPSK, CPFSK, and GFSK. The Fig. 6 shows the confusion matrix of test data split into different SNRs. The

significant aspect observed is that the confusion matrix makes an almost perfect diagonal at high SNR values. It can be noted from the figure that the algorithm is confused at -8 dB SNR, but as the SNR increases, the classification is almost perfect. Furthermore, Fig. 6d shows the best classification result obtained at 18 dB SNR. However, the algorithm is still a bit confused about the classification of WBFM and 8PSK.

The proper way to analyze the results is to break down the test model's accuracy across different SNR values and generate the efficiency of the algorithm for each SNR. The highest accuracy obtained by the proposed model is 96.57% at 18 dB SNR. Moreover, the general trend observed in the results is that the accuracy value increases significantly from 20.59% at -20 dB to 96.57% at 18 dB.

The Fig. 7 indicates the trend observed in test accuracy at different SNR values.

The Gaussian noise and dropout layer acts as a regularization method to improve the model's robustness and efficiency. The system described in this paper was tested with different

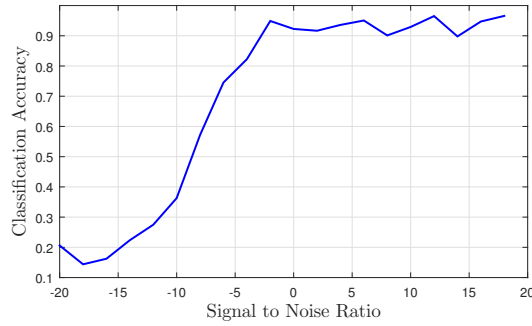


Fig. 7: Classification accuracy v/s SNR plot for the best model

TABLE II: ACCURACY COMPARISON TABLE

Model	Test Accuracy	loss	Accuracy at 18 dB SNR
ANN [11]	37.47%	1.680	51.91%
VTCNN2 [10]	62.59%	0.9580	91.91%
Seven layer CNN [13]	63.04%	0.9693	95.64%
CNN [12]	65.39%	0.9105	94.33%
CNN [14]	62.89%	0.9691	94.03%
Proposed Model	68.94%	0.8242	97.24%

parameters (the standard deviation in the case of Gaussian noise and rate in case of a dropout layer). The variation in parameters generated different results, which were evaluated by checking the accuracy versus SNR plot.

The accuracy of the proposed algorithm is tested for different values of Gaussian noise parameters. The highest accuracy of 97.24% was observed at 0.0001 standard deviation value in the Gaussian noise layer and 18dB SNR. Furthermore, it is observed that, by proper selection of regularization parameters, we can significantly improve the accuracy of the model.

The Table II shows a brief comparison of results of all the CNN based algorithms. It can be seen that the proposed architecture performs better than other models. Our algorithm also outperforms recently proposed CNN based algorithm in [14]. Also, the proposed algorithm is having reduced complexity compared to the one proposed in [14].

VI. CONCLUSION

In this work, we propose a novel CNN based deep learning architecture for automatic modulation classification. The main focus was to obtain high classification accuracy for the test dataset and minimize the validation loss. The proposed architecture provides promising performance in the classification of the test dataset and minimization the validation loss. Moreover, this work also highlights the benefit of using Gaussian noise layer after the flatten layer to improve the regularization performance and reduce the over-fitting. The proposed architecture also outperforms the state-of-the-art CNN based architectures.

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